Daniel Ku

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Education

University of Illinois at Urbana-Champaign
Master of Computer Science — *Jan 2025 – Present*

University of Illinois at Urbana-Champaign — GPA: 3.81/4.0
Bachelor of Science in Computer Science with Honors — *Aug 2021 – Dec 2024*

Experience

Aeyesafe — Remote, USA
Full-stack Intern — *October 2025 – January 2026*

- Engineered a fault-tolerant data ingestion pipeline for HC2 health monitoring sensors, securing the continuous data streaming, webhook event processing, and real-time alerting essential for reliable resident tracking
- Enhanced application performance by diagnosing and eliminating React inefficiencies and refactored legacy JavaScript/CSS into modular components ensuring a responsive, lag-free interface for critical monitoring

Luca Park Share — Taipei, Taiwan
Software Developer — *May 2025 – October 2025*

- Designed the database schema for reservations and payments, implementing atomic transactions to prevent booking conflicts and ensure system stability
- Implemented booking features focusing on performance, optimizing SQL queries for fast responses and real-time availability checks. Built the admin dashboard to monitor system usage for tracking application health and user metrics

Steel Perlot — NYC, New York
Research Intern — *May 2022 – Aug 2022*

- Launched a startup incubation program providing $1.5M in grants to builders working on Web3 problems
- Networked and organized satellite events at Web3 conferences and hackathons to scout potential hires and investments
- Conducted technical analysis on potential startup investments, market sentiment, and various competitive landscapes

Projects

IoT Bicycle Hazard Detection Project (CS 437) | *C++, Python* — Jan 2024 – May 2024

- Built a low-latency hazard detection system on embedded hardware (Raspberry Pi + Coral TPU), optimizing real-time vision inference and demonstrating end-to-end sensor to processing to alert pipelines
- Developed a BLE-connected smart taillight on ESP32-S3, optimizing Bluetooth throughput, latency, and reconnection stability, driving a real-time 8×8 LED alert system with multiprocessing queues

Rotarygolf.info | *React, TypeScript, PostgreSQL* — September 2025 – Present

- Developed a full-stack golf tournament management system using React, TypeScript, and PostgreSQL serving a 50-member golf club with plans to scale to ∼100 clubs in Taipei
- Utilized fast text recognition OCR models to collect golf score data, created an intuitive UI designed for easy adoption by non-technical users, featuring role-based authentication, real-time leaderboards, automated award calculations

Numerical Methods for Shallow-Water PDEs Project (CS 555) | *Python* — Jan 2025 – May 2025

- Ensured algorithmic efficiency in finite-volume methods for shallow-water PDEs; optimized timestep selection, memory layout, and computational throughput for numerically stable simulations
- Analyzed solution stability and mesh-dependent behavior by comparing triangular vs. uniform grid performance, developed visualizations in Python to detect instabilities and validate correctness under varying numerical conditions

Technical Skills

Programming: Python, TypeScript, Java, C++, Rust
DevOps & Cloud: Cursor MCP, Docker, AWS, Railway
Frameworks & Libraries: Tensorflow.js, WebSockets
Databases: PostgreSQL, MongoDB